

REVIEW OF (LAST PART OF) THE PREVIOUS LECTURE

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Sample Mean:

$X_1, \dots, X_n$ iid with mean μ and variance σ^2 . $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$

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Chebyshev's Applied on $\bar{X}_n$: $P(|\bar{X}_n - \mu| \geq t) \leq \frac{\sigma^2}{nt^2}$

READINGS ON THE LAST LECTURE AND THE NEXT FEW LECTURES

10.1, 11.2, 11.2, 11.5, Ch 12, 14.1, 14.2, 14.3, 15.1-15.5
[Watkins]

4.1 (Markov's, Chebyshev's Inequalities),

5.1, 5.3, 5.4, Ch 6, 9.1, 9.3-9.5
[Wasserman]

SAMPLE MEAN

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$\bar{X}_n$ has mean p , variance $\leq \frac{1/4}{100,000}$. Chebyshev's Inequality on $\bar{X}_n$:

$$P(|\bar{X}_n - p| \geq 0.01) \leq \frac{1/4}{0.01^2 \times 100,000} = 2.5 \%$$

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$\bar{X}_n$ has mean μ and variance $\frac{8}{n}$. Chebyshev's Inequality on $\bar{X}_n$:

$$P(|\bar{X}_n - \mu| < 2) \geq 1 - \frac{8}{4n} = 1 - \frac{2}{n}$$

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$$1 - \frac{2}{n} = 0.99 \Rightarrow 0.01n = 2 \Rightarrow n \text{ at least } 200.$$

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Formally: Let $X_1, \dots, X_n$ iid with mean μ , standard deviation σ

$$P(|\bar{X}_n - \mu| < t) \geq 1 - \frac{\sigma^2}{nt^2}, \quad \text{for any } t > 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| < t) = 1$$

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$$\text{As } n \rightarrow \infty \quad Z_n = \frac{\bar{X}_n - \mu}{\sqrt{\text{Var}(\bar{X}_n)}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \sim N(0,1)$$

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$$\Rightarrow P(\bar{X}_n < 4.2) = P\left(\frac{\sqrt{100}(\bar{X}_n - 4)}{\sqrt{4}} < \frac{\sqrt{100}(4.2 - 4)}{\sqrt{4}}\right)$$

$$\approx P(Z < 1) = 0.84$$

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$$\begin{aligned} \Rightarrow P(\bar{X}_n \leq 72) &= P\left(\frac{\sqrt{100}(\bar{X}_n - 80)}{40} \leq \frac{\sqrt{100}(72 - 80)}{40}\right) \\ &\approx P(Z \leq -2) \approx 0.023 \end{aligned}$$